

3.8.4 Gene technologies allow the study and alteration of gene function allowing a better understanding of organism function and the design of new industrial and medical processes (A-level only)

3.8.4.1 Recombinant DNA technology (A-level only)

Content	Opportunities for skills development
Recombinant DNA technology involves the transfer of fragments of DNA from one organism, or species, to another. Since the genetic code is universal, as are transcription and translation mechanisms, the transferred DNA can be translated within cells of the recipient (transgenic) organism. Fragments of DNA can be produced by several methods, including: • conversion of mRNA to complementary DNA (cDNA), using reverse transcriptase • using restriction enzymes to cut a fragment containing the desired gene from DNA • creating the gene in a 'gene machine'. Fragments of DNA can be amplified by <i>in vitro</i> and <i>in vivo</i> techniques. The principles of the polymerase chain reaction (PCR) as an <i>in vitro</i> method to amplify DNA fragments. The culture of transformed host cells as an <i>in vivo</i> method to amplify DNA fragments. • The addition of promoter and terminator regions to the fragments of DNA. • The use of restriction endonucleases and ligases to insert fragments of DNA into vectors. Transformation of host cells using these vectors. • The use of marker genes to detect genetically modified (GM) cells or organisms. (Students will not be required to recall specific marker genes in a written paper.) Students should be able to: • interpret information relating to the use of recombinant DNA technology • evaluate the ethical, financial and social issues associated with the use and ownership of recombinant DNA technology in agriculture, in industry and in medicine • balance the humanitarian aspects of recombinant DNA technology with the opposition from environmentalists and anti-globalisation activists • relate recombinant DNA technology to gene therapy.	AT g Students could investigate the specificity of restriction enzymes using extracted DNA and electrophoresis.

3.8.4.2 Differences in DNA between individuals of the same species can be exploited for identification and diagnosis of heritable conditions (A-level only)

Content	Opportunities for skills development
The use of labelled DNA probes and DNA hybridisation to locate specific alleles of genes. The use of labelled DNA probes that can be used to screen patients for heritable conditions, drug responses or health risks. The use of this information in genetic counselling and personalised medicine. Students should be able to evaluate information relating to screening individuals for genetically determined conditions and drug responses.	

3.8.4.3 Genetic fingerprinting (A-level only)

Content	Opportunities for skills development
An organism's genome contains many variable number tandem repeats (VNTRs). The probability of two individuals having the same VNTRs is very low. The technique of genetic fingerprinting in analysing DNA fragments that have been cloned by PCR, and its use in determining genetic relationships and in determining the genetic variability within a population. The use of genetic fingerprinting in the fields of forensic science, medical diagnosis, animal and plant breeding. Students should be able to: • explain the biological principles that underpin genetic fingerprinting techniques • interpret data showing the results of gel electrophoresis to separate DNA fragments • explain why scientists might use genetic fingerprinting in the fields of forensic science, medical diagnosis, animal and plant breeding.	AT g Students could use gel electrophoresis to produce 'fingerprints' of food dyes.